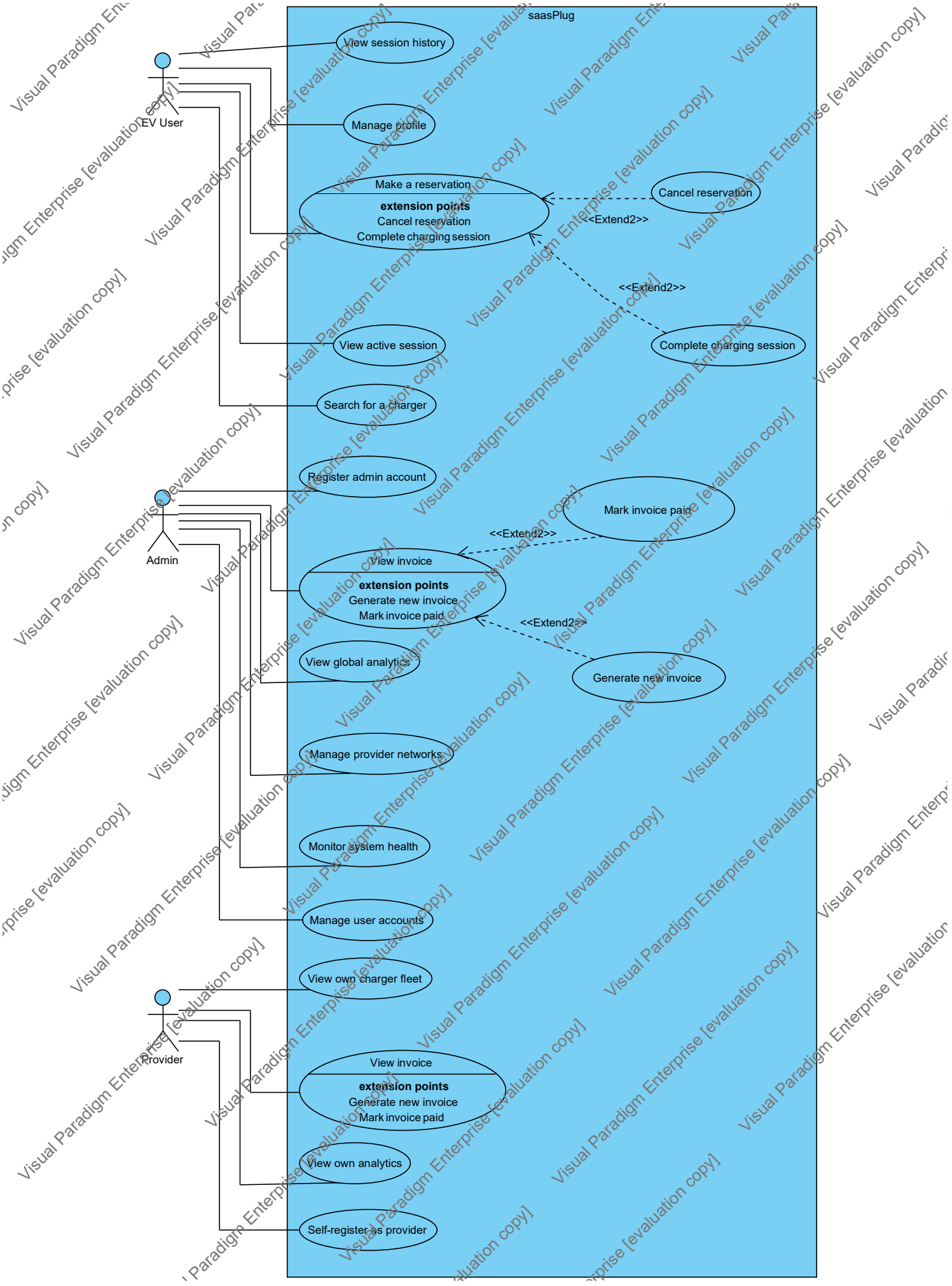
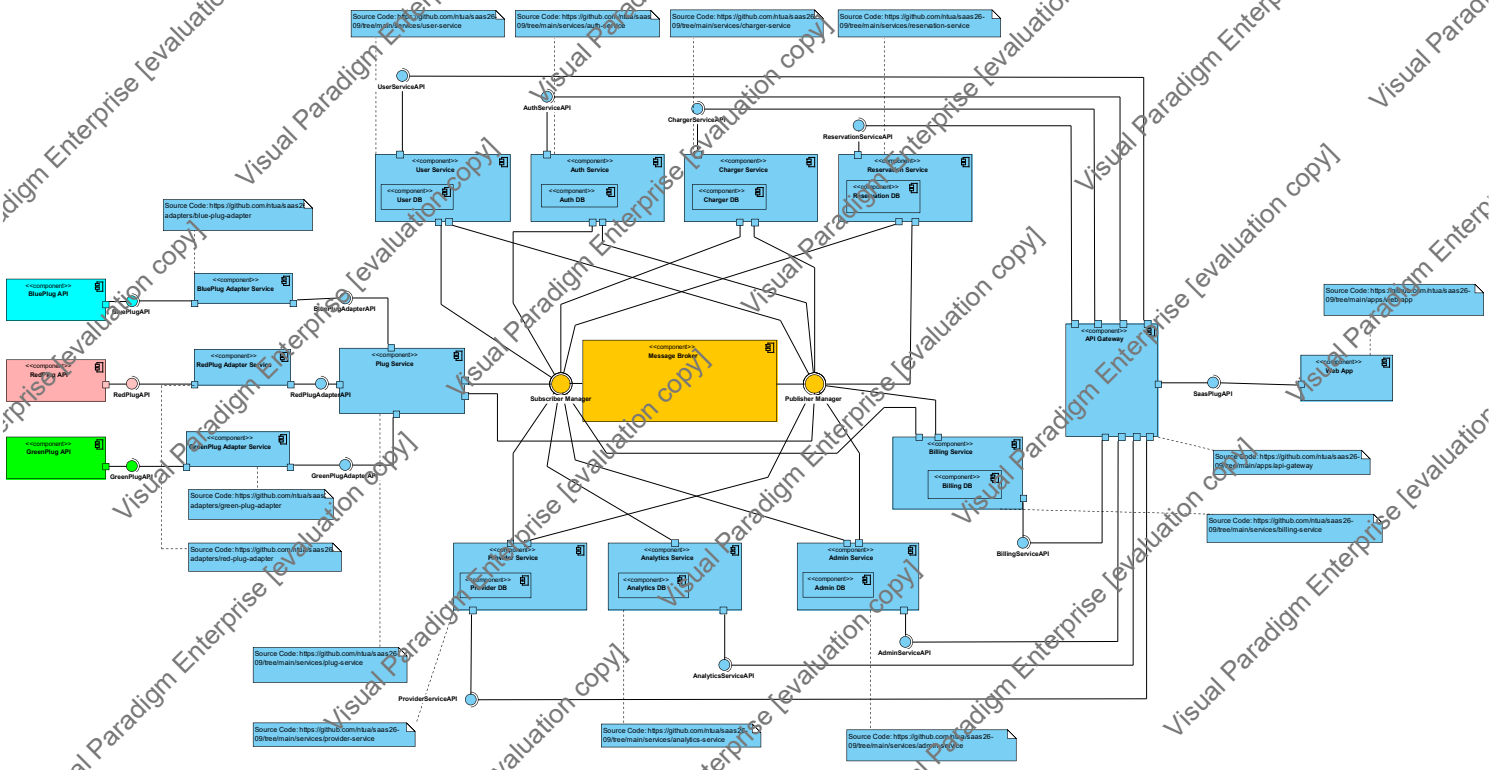
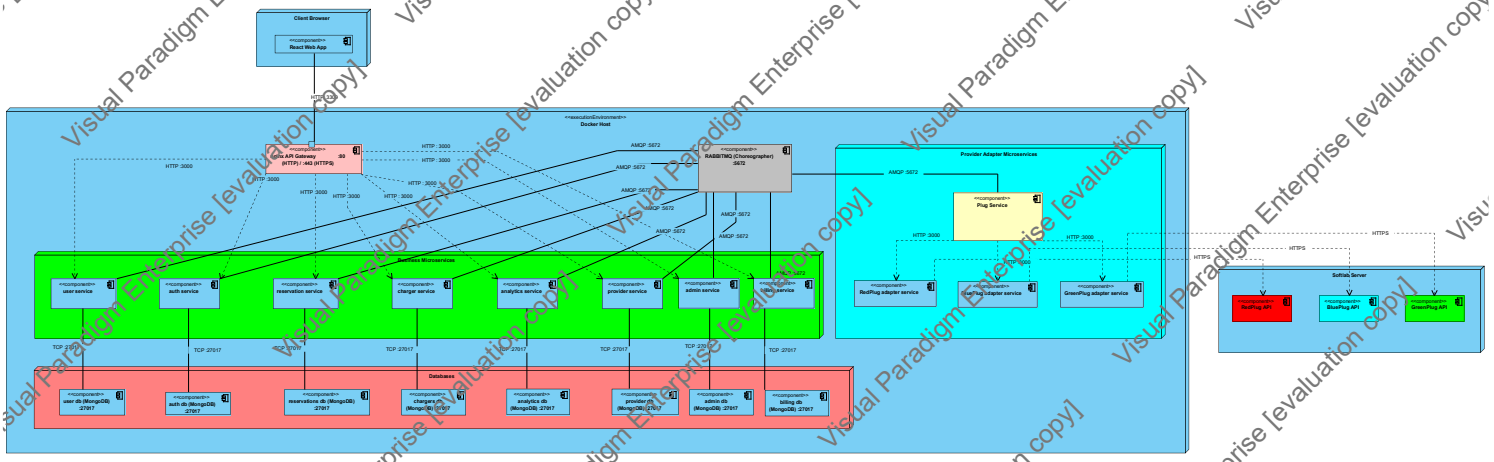
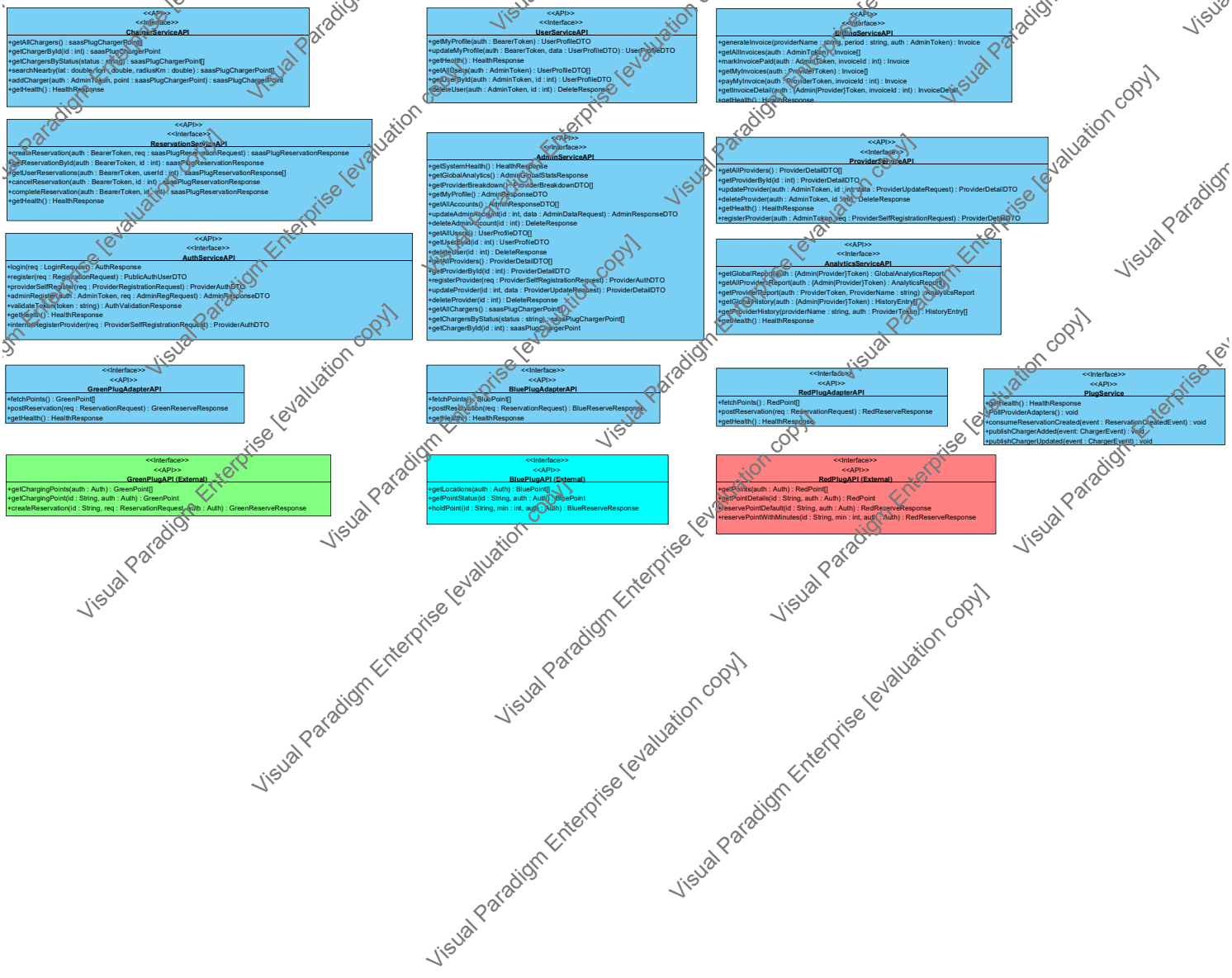
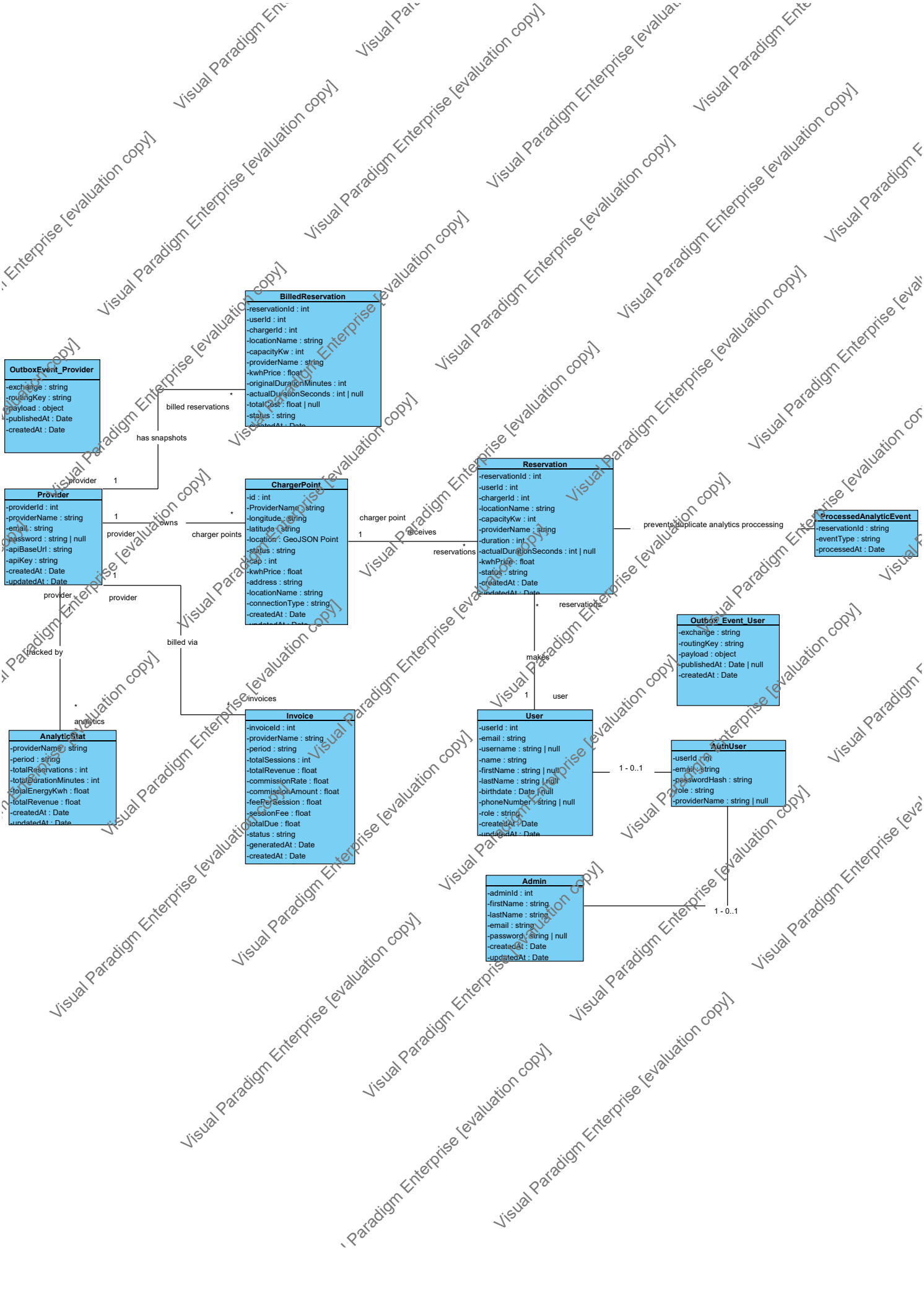

saasPlug
EV User
View session history
Manage profile
Make a reservation
extension points
Cancel reservation
Complete charging session
Cancel reservation
<<Extend2>>
<<Extend2>>
Complete charging session
View active session
Search for a charger
Admin
Register admin account
Mark invoice paid
<<Extend2>>
View invoice
extension points
Generate new invoice
Mark invoice paid
<<Extend2>>
Generate new invoice
View global analytics
Manage provider networks
Monitor system health
Manage user accounts
View own charger fleet
Provider
View invoice
extension points
Generate new invoice
Mark invoice paid
View own analytics
Self-register as provider

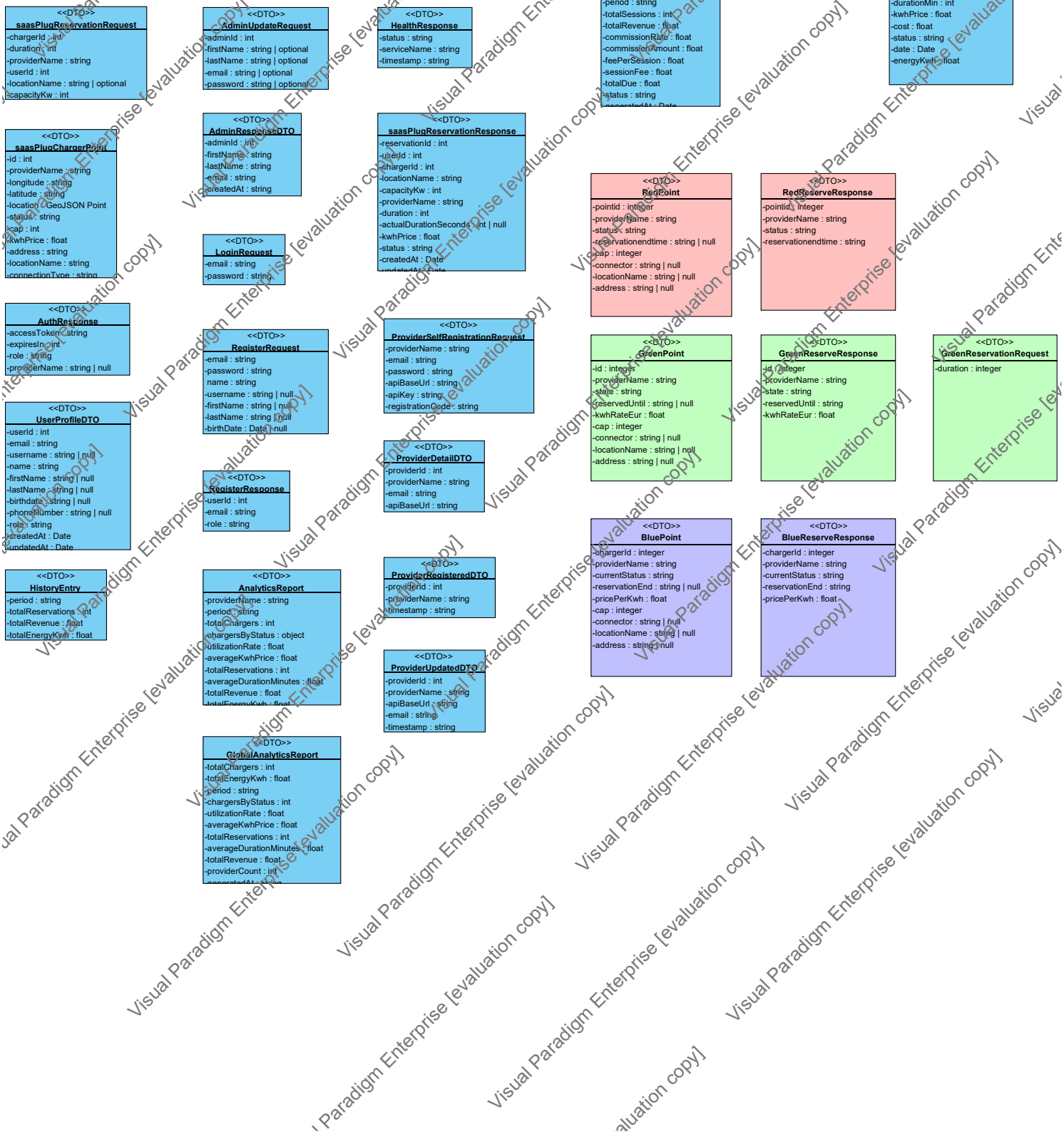

Source Code: https://github.com/ntua/saas26-09/tree/main/services/user-service
Source Code: https://github.com/ntua/saas26-09/tree/main/services/auth-service
Source Code: https://github.com/ntua/saas26-09/tree/main/services/charger-service
Source Code: https://github.com/ntua/saas26-09/tree/main/services/reservation-service
UserServiceAPI
AuthServiceAPI
ChargerServiceAPI
ReservationServiceAPI
User Service
User DB
Auth Service
Auth DB
Charger Service
Charger DB
Reservation Service
Reservation DB
Source Code: https://github.com/ntua/saas26-09/tree/main/adapters/blue-plug-adapter
BluePlug API
BluePlugAPI
BluePlug Adapter Service
BluePlugAdapterAPI
RedPlug API
RedPlugAPI
RedPlug Adapter Service
RedPlugAdapterAPI
Plug Service
Message Broker
Subscriber Manager
Publisher Manager
API Gateway
SaasPlugAPI
Source Code: https://github.com/ntua/saas26-09/tree/main/apps/web-app
Web App
GreenPlug API
GreenPlugAPI
GreenPlug Adapter Service
GreenPlugAdapterAPI
Source Code: https://github.com/ntua/saas26-09/tree/main/adapters/green-plug-adapter
Source Code: https://github.com/ntua/saas26-09/tree/main/adapters/red-plug-adapter
Billing Service
Billing DB
BillingServiceAPI
Source Code: https://github.com/ntua/saas26-09/tree/main/apps/api-gateway
Source Code: https://github.com/ntua/saas26-09/tree/main/services/billing-service
Provider Service
Provider DB
Analytics Service
Analytics DB
Admin Service
Admin DB
AdminServiceAPI
AnalyticsServiceAPI
ProviderServiceAPI
Source Code: https://github.com/ntua/saas26-09/tree/main/services/plug-service
Source Code: https://github.com/ntua/saas26-09/tree/main/services/provider-service
Source Code: https://github.com/ntua/saas26-09/tree/main/services/analytics-service
Source Code: https://github.com/ntua/saas26-09/tree/main/services/admin-service

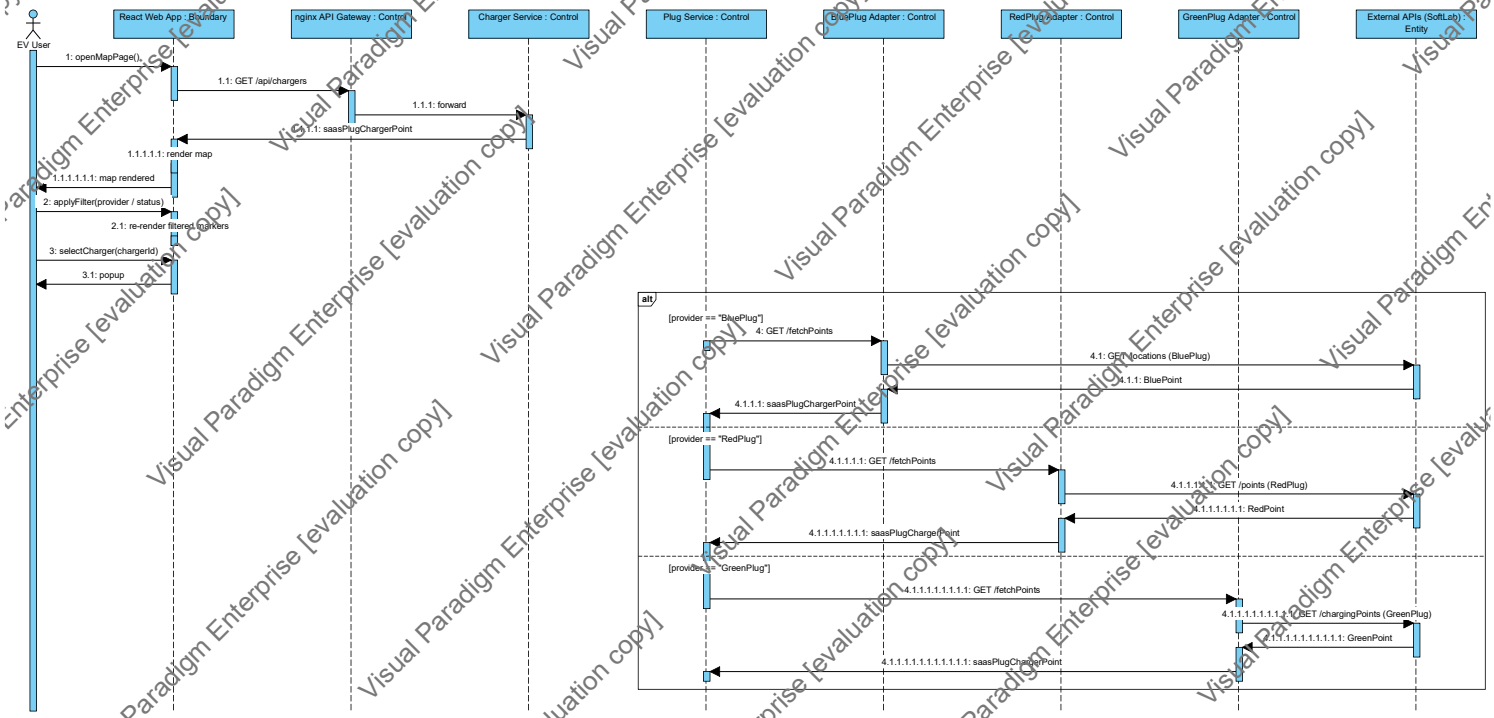


Visual Paradigm Enterprise [evaluation copy]

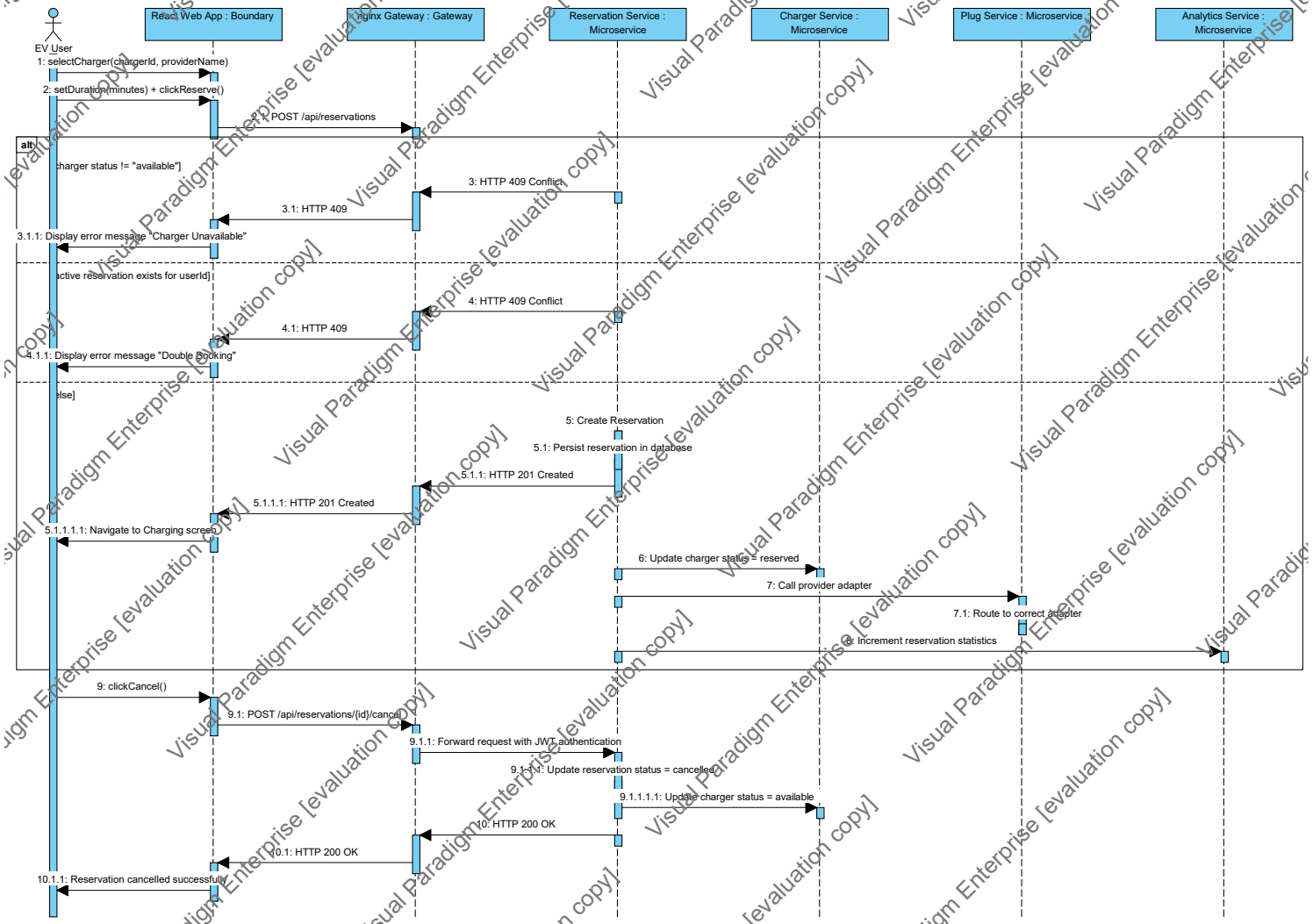

<<API>>
<<Interface>>
ChargerServiceAPI
+getAllChargers() : saasPlugChargerPoint[]
+getChargerById(id : int) : saasPlugChargerPoint
+getChargersByStatus(status : string) : saasPlugChargerPoint[]
+searchNearby(lat : double, lon : double, radiusKm : double) : saasPlugChargerPoint[]
+addCharger(auth : AdminToken, point : saasPlugChargerPoint) : saasPlugChargerPoint
+getHealth() : HealthResponse
<<API>>
<<Interface>>
UserServiceAPI
+getMyProfile(auth : BearerToken) : UserProfileDTO
+updateMyProfile(auth : BearerToken, data : UserProfileDTO) : UserProfileDTO
+getHealth() : HealthResponse
+getAllUsers(auth : AdminToken) : UserProfileDTO[]
+getUserById(auth : AdminToken, id : int) : UserProfileDTO
+deleteUser(auth : AdminToken, id : int) : DeleteResponse
<<API>>
<<Interface>>
BillingServiceAPI
+generateInvoice(providerName : string, period : string, auth : AdminToken) : Invoice
+getAllInvoices(auth : AdminToken) : Invoice[]
+markInvoicePaid(auth : AdminToken, invoiceId : int) : Invoice
+getMyInvoices(auth : ProviderToken) : Invoice[]
+payMyInvoice(auth : ProviderToken, invoiceId : int) : Invoice
+getInvoiceDetail(auth : [Admin|Provider]Token, invoiceId : int) : InvoiceDetail
+getHealth() : HealthResponse
<<API>>
<<Interface>>
ReservationServiceAPI
+createReservation(auth : BearerToken, req : saasPlugReservationRequest) : saasPlugReservationResponse
+getReservationById(auth : BearerToken, id : int) : saasPlugReservationResponse
+getUserReservations(auth : BearerToken, userId : int) : saasPlugReservationResponse[]
+cancelReservation(auth : BearerToken, id : int) : saasPlugReservationResponse
+completeReservation(auth : BearerToken, id : int) : saasPlugReservationResponse
+getHealth() : HealthResponse
<<API>>
<<Interface>>
AdminServiceAPI
+getSystemHealth() : HealthResponse
+getGlobalAnalytics() : AdminGlobalStatsResponse
+getProviderBreakdown() : ProviderBreakdownDTO[]
+getMyProfile() : AdminResponseDTO
+getAllAccounts() : AdminResponseDTO[]
+updateAdminAccount(id : int, data : AdminDataRequest) : AdminResponseDTO
+deleteAdminAccount(id : int) : DeleteResponse
+getAllUsers() : UserProfileDTO[]
+getUserById(id : int) : UserProfileDTO
+deleteUser(id : int) : DeleteResponse
+getAllProviders() : ProviderDetailDTO[]
+getProviderById(id : int) : ProviderDetailDTO
+registerProvider(req : ProviderSelfRegistrationRequest) : ProviderAuthDTO
+updateProvider(id : int, data : ProviderUpdateRequest) : ProviderDetailDTO
+deleteProvider(id : int) : DeleteResponse
+getAllChargers() : saasPlugChargerPoint[]
+getChargersByStatus(status : string) : saasPlugChargerPoint[]
+getChargerById(id : int) : saasPlugChargerPoint
<<API>>
<<Interface>>
ProviderServiceAPI
+getAllProviders() : ProviderDetailDTO[]
+getProviderById(id : int) : ProviderDetailDTO
+updateProvider(auth : AdminToken, id : int, data : ProviderUpdateRequest) : ProviderDetailDTO
+deleteProvider(auth : AdminToken, id : int) : DeleteResponse
+getHealth() : HealthResponse
+registerProvider(auth : AdminToken, req : ProviderSelfRegistrationRequest) : ProviderDetailDTO
<<API>>
<<Interface>>
AuthServiceAPI
+login(req : LoginRequest) : AuthResponse
+register(req : RegistrationRequest) : PublicAuthUserDTO
+providerSelfRegister(req : ProviderRegistrationRequest) : ProviderAuthDTO
+adminRegister(auth : AdminToken, req : AdminRegRequest) : AdminResponseDTO
+validateToken(token : string) : AuthValidationResponse
+getHealth() : HealthResponse
+internalRegisterProvider(req : ProviderSelfRegistrationRequest) : ProviderAuthDTO
<<API>>
<<Interface>>
AnalyticsServiceAPI
+getGlobalReport(auth : [Admin|Provider]Token) : GlobalAnalyticsReport
+getAllProvidersReport(auth : [Admin|Provider]Token) : AnalyticsReport[]
+getProviderReport(auth : ProviderToken, ProviderName : string) : AnalyticsReport
+getGlobalHistory(auth : [Admin|Provider]Token) : HistoryEntry[]
+getProviderHistory(providerName : string, auth : ProviderToken) : HistoryEntry[]
+getHealth() : HealthResponse
<<Interface>>
<<API>>
GreenPlugAdapterAPI
+fetchPoints() : GreenPoint[]
+postReservation(req : ReservationRequest) : GreenReserveResponse
+getHealth() : HealthResponse
<<Interface>>
<<API>>
BluePlugAdapterAPI
+fetchPoints() : BluePoint[]
+postReservation(req : ReservationRequest) : BlueReserveResponse
+getHealth() : HealthResponse
<<Interface>>
<<API>>
RedPlugAdapterAPI
+fetchPoints() : RedPoint[]
+postReservation(req : ReservationRequest) : RedReserveResponse
+getHealth() : HealthResponse
<<Interface>>
<<API>>
PlugService
+getHealth() : HealthResponse
+postProviderAdapters() : void
+consumeReservationCreated(event : ReservationCreatedEvent) : void
+publishChargerAdded(event: ChargerEvent) : void
+publishChargerUpdated(event : ChargerEvent) : void
<<Interface>>
<<API>>
GreenPlugAPI (External)
+getChargingPoints(auth : Auth) : GreenPoint[]
+getChargingPoint(id : String, auth : Auth) : GreenPoint
+createReservation(id : String, req : ReservationRequest, auth : Auth) : GreenReserveResponse
<<Interface>>
<<API>>
BluePlugAPI (External)
+getLocations(auth : Auth) : BluePoint[]
+getPointStatus(id : String, auth : Auth) : BluePoint
+holdPoint(id : String, min : int, auth : Auth) : BlueReserveResponse
<<Interface>>
<<API>>
RedPlugAPI (External)
+getPoints(auth : Auth) : RedPoint[]
+getPointDetails(id : String, auth : Auth) : RedPoint
+reservePointDefault(id : String, auth : Auth) : RedReserveResponse
+reservePointWithMinutes(id : String, min : int, auth : Auth) : RedReserveResponse

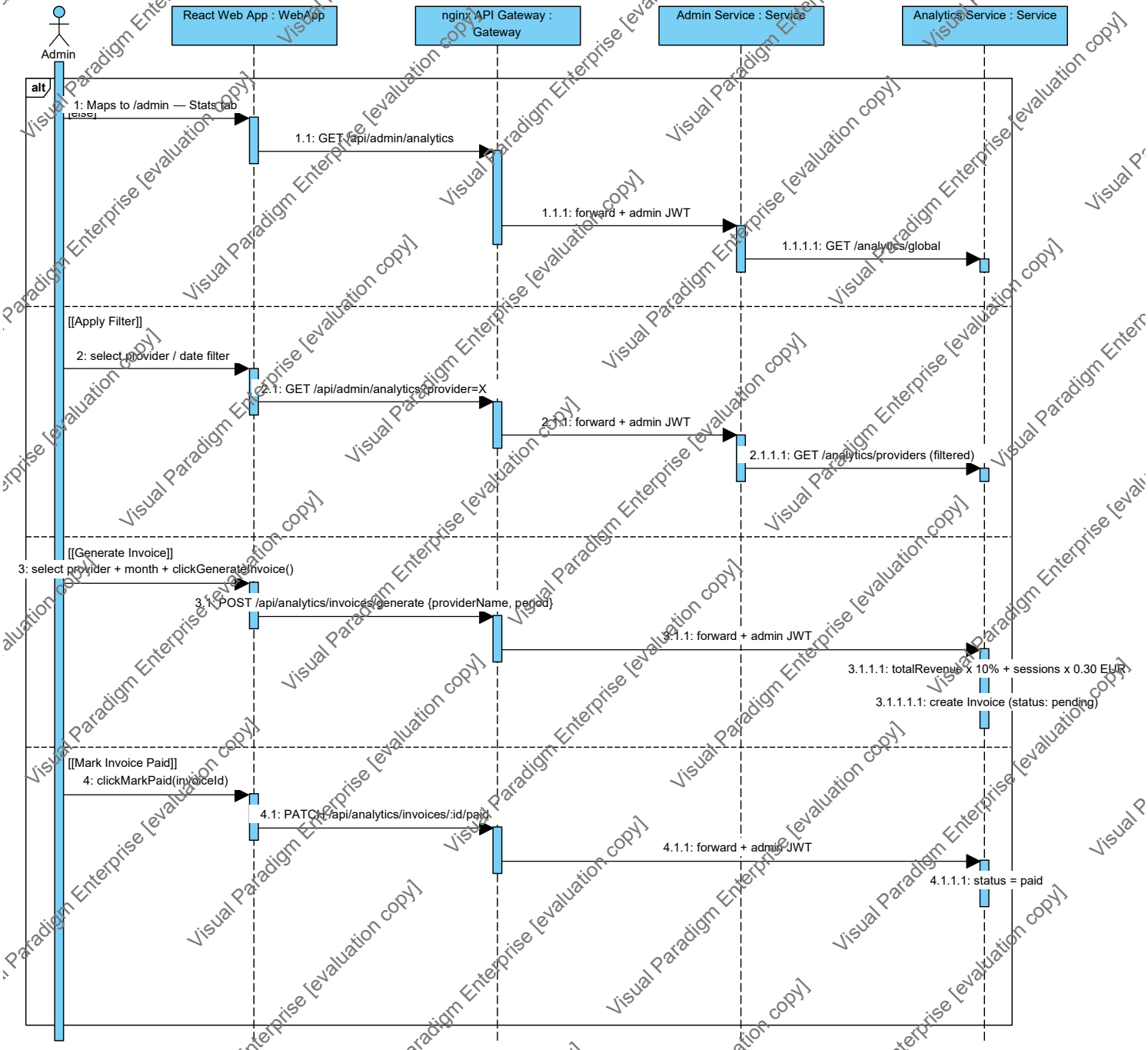

BilledReservation
-reservationId : int
-userId : int
-chargerId : int
-locationName : string
-capacityKw : int
-providerName : string
-kwhPrice : float
-originalDurationMinutes : int
-actualDurationSeconds : int | null
-totalCost : float | null
-status : string
OutboxEvent_Provider
-exchange : string
-routingKey : string
-payload : object
-publishedAt : Date
-createdAt : Date
billed reservations
has snapshots
provider
Provider
-providerId : int
-providerName : string
-email : string
-password : string | null
-apiBaseUrl : string
-apiKey : string
-createdAt : Date
-updatedAt : Date
owns
charger points
ChargerPoint
-id : int
-ProviderName : string
-longitude : string
-latitude : string
-location : GeoJSON Point
-status : string
-cap : int
-kwhPrice : float
-address : string
-locationName : string
-connectionType : string
-createdAt : Date
charger point
receives
reservations
Reservation
-reservationId : int
-userId : int
-chargerId : int
-locationName : string
-capacityKw : int
-providerName : string
-duration : int
-actualDurationSeconds : int | null
-kwhPrice : float
-status : string
-createdAt : Date
prevents duplicate analytics proccessing
ProcessedAnalyticEvent
-reservationId : string
-eventType : string
-processedAt : Date
tracked by
analytics
AnalyticStat
-providerName : string
-period : string
-totalReservations : int
-totalDurationMinutes : int
-totalEnergyKwh : float
-totalRevenue : float
-createdAt : Date
-updatedAt : Date
billed via
invoices
Invoice
-invoiceId : int
-providerName : string
-period : string
-totalSessions : int
-totalRevenue : float
-commissionRate : float
-commissionAmount : float
-feePerSession : float
-sessionFee : float
-totalDue : float
-status : string
-generatedAt : Date
-createdAt : Date
makes
user
User
-userId : int
-email : string
-username : string | null
-name : string
-firstName : string | null
-lastName : string | null
-birthdate : Date | null
-phoneNumber : string | null
-role : string
-createdAt : Date
-updatedAt : Date
Outbox_Event_User
-exchange : string
-routingKey : string
-payload : object
-publishedAt : Date | null
-createdAt : Date
1 - 0..1
AuthUser
-userId : int
-email : string
-passwordHash : string
-role : string
-providerName : string | null
Admin
-adminId : int
-firstName : string
-lastName : string
-email : string
-password : string | null
-createdAt : Date
-updatedAt : Date
1 - 0..1

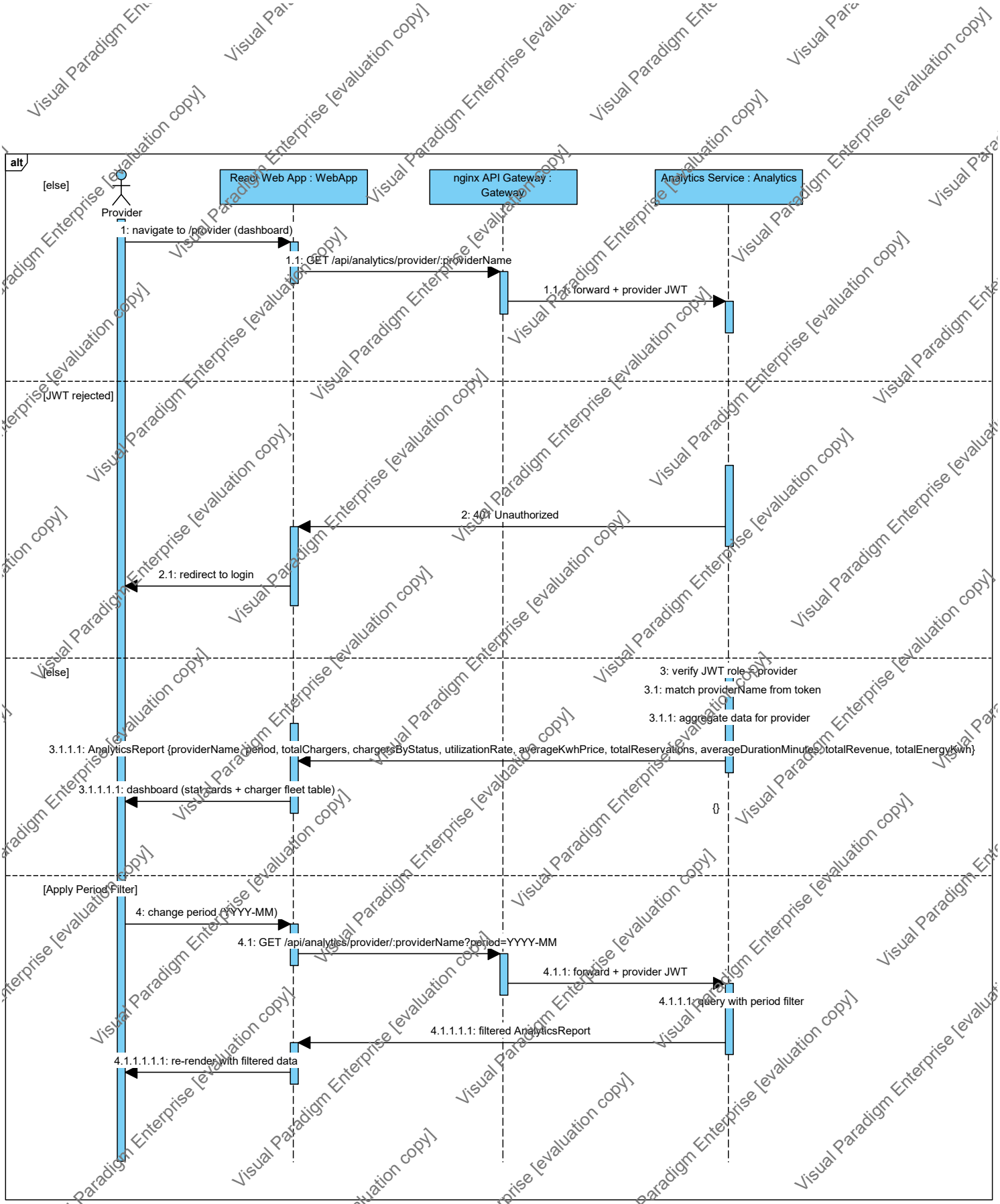

<<DTO>>
saasPlugReservationRequest
-chargerId : int
-duration : int
-providerName : string
-userId : int
-locationName : string | optional
-capacityKw : int
<<DTO>>
AdminUpdateRequest
-adminId : int
-firstName : string | optional
-lastName : string | optional
-email : string | optional
-password : string | optional
<<DTO>>
HealthResponse
-status : string
-serviceName : string
-timestamp : string
<<DTO>>
InvoiceDTO
-invoiceId : int
-providerName : string
-period : string
-totalSessions : int
-totalRevenue : float
-commissionRate : float
-commissionAmount : float
-feePerSession : float
-sessionFee : float
-totalDue : float
-status : string
<<DTO>>
InvoiceDetailDTO
-invoice : InvoiceDTO
-lines : ReservationLine[]
<<DTO>>
ReservationLine
-reservationId : int
-locationName : string
-durationMin : int
-kwhPrice : float
-cost : float
-status : string
-date : Date
-energyKwh : float
<<DTO>>
saasPlugChargerPoint
-id : int
-providerName : string
-longitude : string
-latitude : string
-location : GeoJSON Point
-status : string
-cap : int
-kwhPrice : float
-address : string
-locationName : string
-connectionType : string
<<DTO>>
AdminResponseDTO
-adminId : int
-firstName : string
-lastName : string
-email : string
-createdAt : string
<<DTO>>
saasPlugReservationResponse
-reservationId : int
-userId : int
-chargerId : int
-locationName : string
-capacityKw : int
-providerName : string
-duration : int
-actualDurationSeconds : int | null
-kwhPrice : float
-status : string
-createdAt : Date
<<DTO>>
RedPoint
-pointId : integer
-providerName : string
-status : string
-reservationendtime : string | null
-cap : integer
-connector : string | null
-locationName : string | null
-address : string | null
<<DTO>>
RedReserveResponse
-pointId : integer
-providerName : string
-status : string
-reservationendtime : string
<<DTO>>
LoginRequest
-email : string
-password : string
<<DTO>>
AuthResponse
-accessToken : string
-expiresIn : int
-role : string
-providerName : string | null
<<DTO>>
RegisterRequest
-email : string
-password : string
name : string
-username : string | null
-firstName : string | null
-lastName : string | null
-birthDate : Date | null
<<DTO>>
ProviderSelfRegistrationRequest
-providerName : string
-email : string
-password : string
-apiBaseUrl : string
-apiKey : string
-registrationCode : string
<<DTO>>
GreenPoint
-id : integer
-providerName : string
-state : string
-reservedUntil : string | null
-kwhRateEur : float
-cap : integer
-connector : string | null
-locationName : string | null
-address : string | null
<<DTO>>
GreenReserveResponse
-id : integer
-providerName : string
-state : string
-reservedUntil : string
-kwhRateEur : float
<<DTO>>
GreenReservationRequest
-duration : integer
<<DTO>>
UserProfileDTO
-userId : int
-email : string
-username : string | null
-name : string
-firstName : string | null
-lastName : string | null
-birthdate : string | null
-phoneNumber : string | null
-role : string
-createdAt : Date
-updatedAt : Date
<<DTO>>
RegisterResponse
-userId : int
-email : string
-role : string
<<DTO>>
ProviderDetailDTO
-providerId : int
-providerName : string
-email : string
-apiBaseUrl : string
<<DTO>>
BluePoint
-chargerId : integer
-providerName : string
-currentStatus : string
-reservationEnd : string | null
-pricePerKwh : float
-cap : integer
-connector : string | null
-locationName : string | null
-address : string | null
<<DTO>>
BlueReserveResponse
-chargerId : integer
-providerName : string
-currentStatus : string
-reservationEnd : string
-pricePerKwh : float
<<DTO>>
HistoryEntry
-period : string
-totalReservations : int
-totalRevenue : float
-totalEnergyKwh : float
<<DTO>>
AnalyticsReport
-providerName : string
-period : string
-totalChargers : int
-chargersByStatus : object
-utilizationRate : float
-averageKwhPrice : float
-totalReservations : int
-averageDurationMinutes : float
-totalRevenue : float
-totalEnergyKwh : float
<<DTO>>
ProviderRegisteredDTO
-providerId : int
-providerName : string
-timestamp : string
<<DTO>>
ProviderUpdatedDTO
-providerId : int
-providerName : string
-apiBaseUrl : string
-email : string
-timestamp : string
<<DTO>>
GlobalAnalyticsReport
-totalChargers : int
-totalEnergyKwh : float
-period : string
-chargersByStatus : int
-utilizationRate : float
-averageKwhPrice : float
-totalReservations : int
-averageDurationMinutes : float
-totalRevenue : float
-providerCount : int

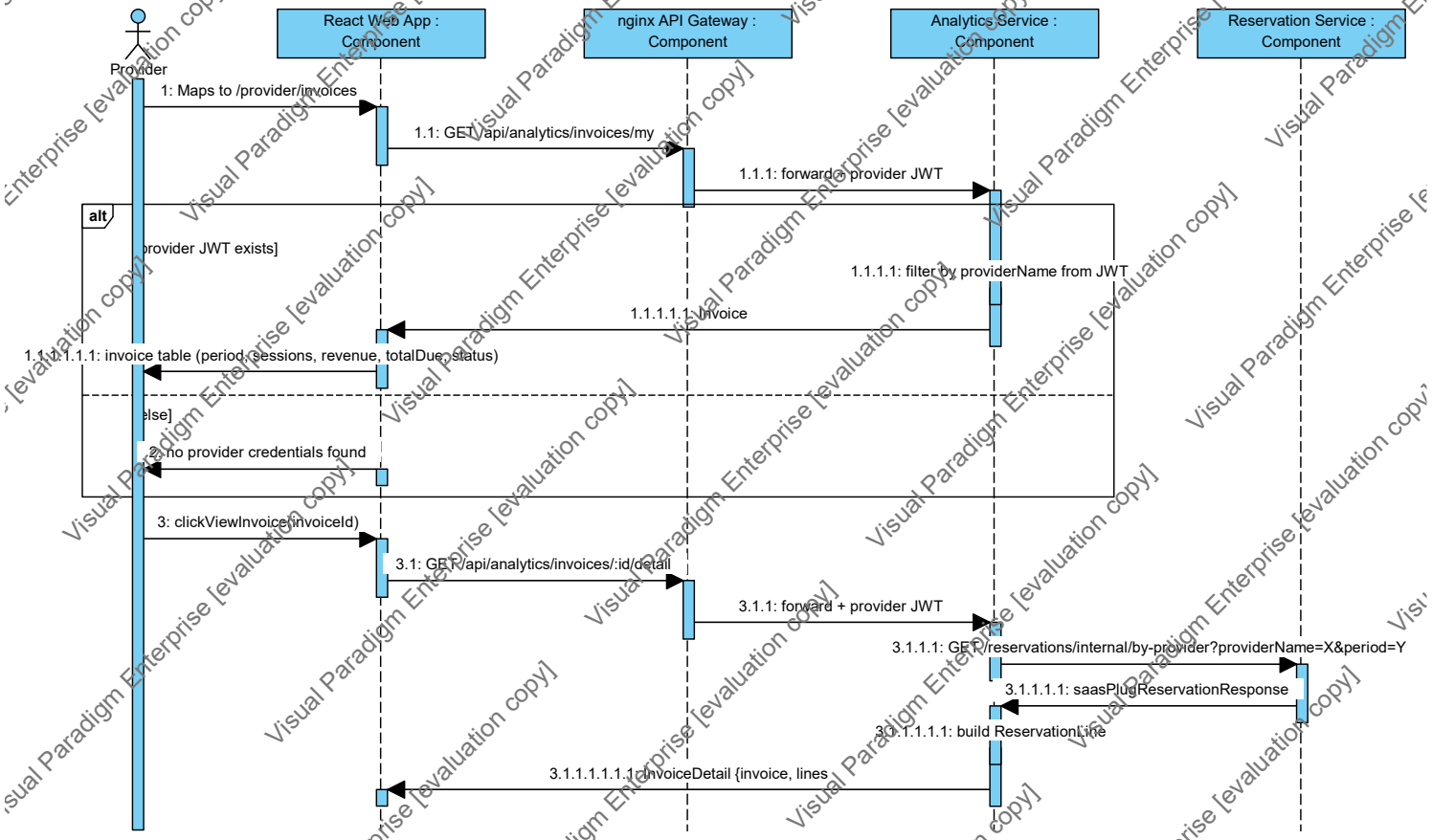


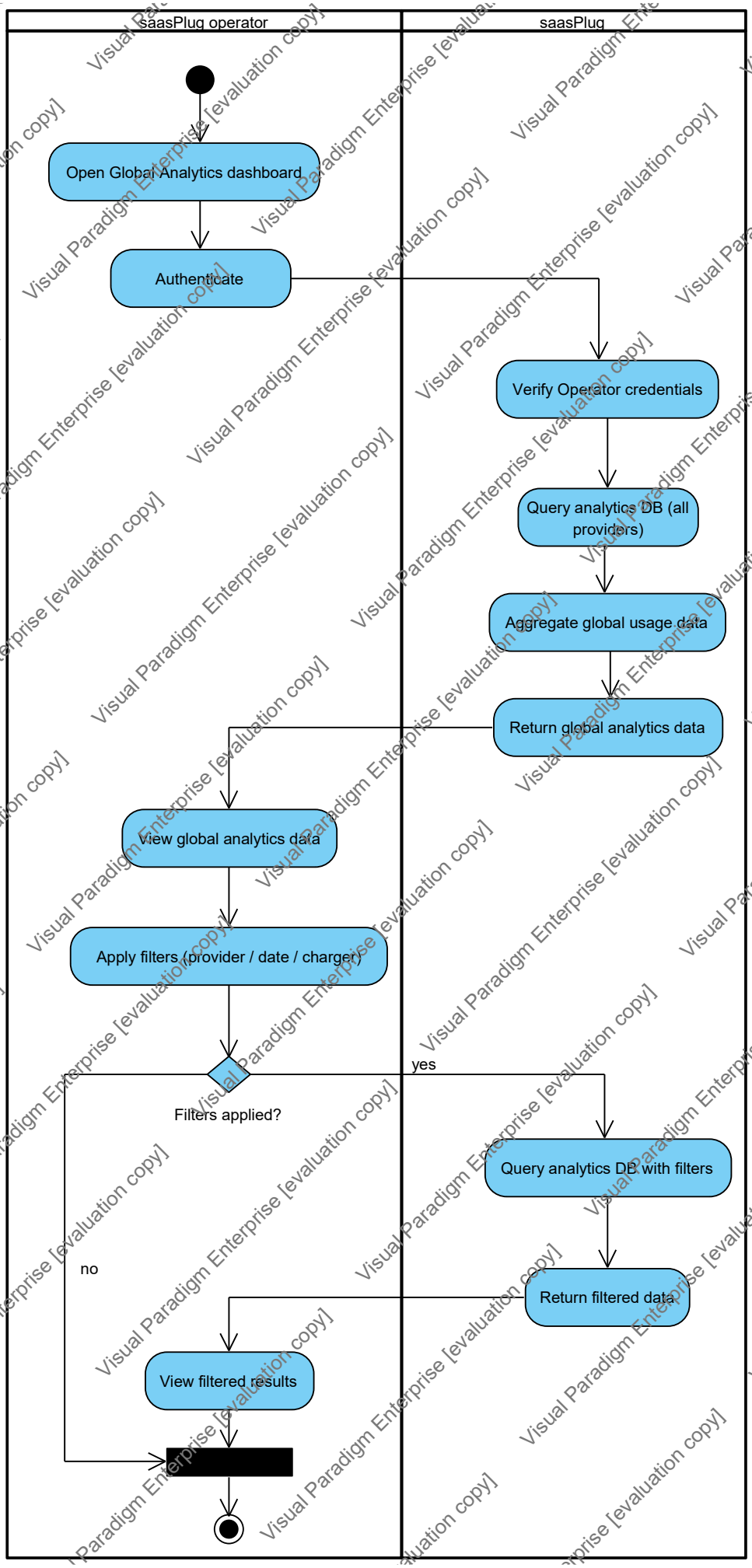
Visual Paradigm Enterprise [evaluation copy]

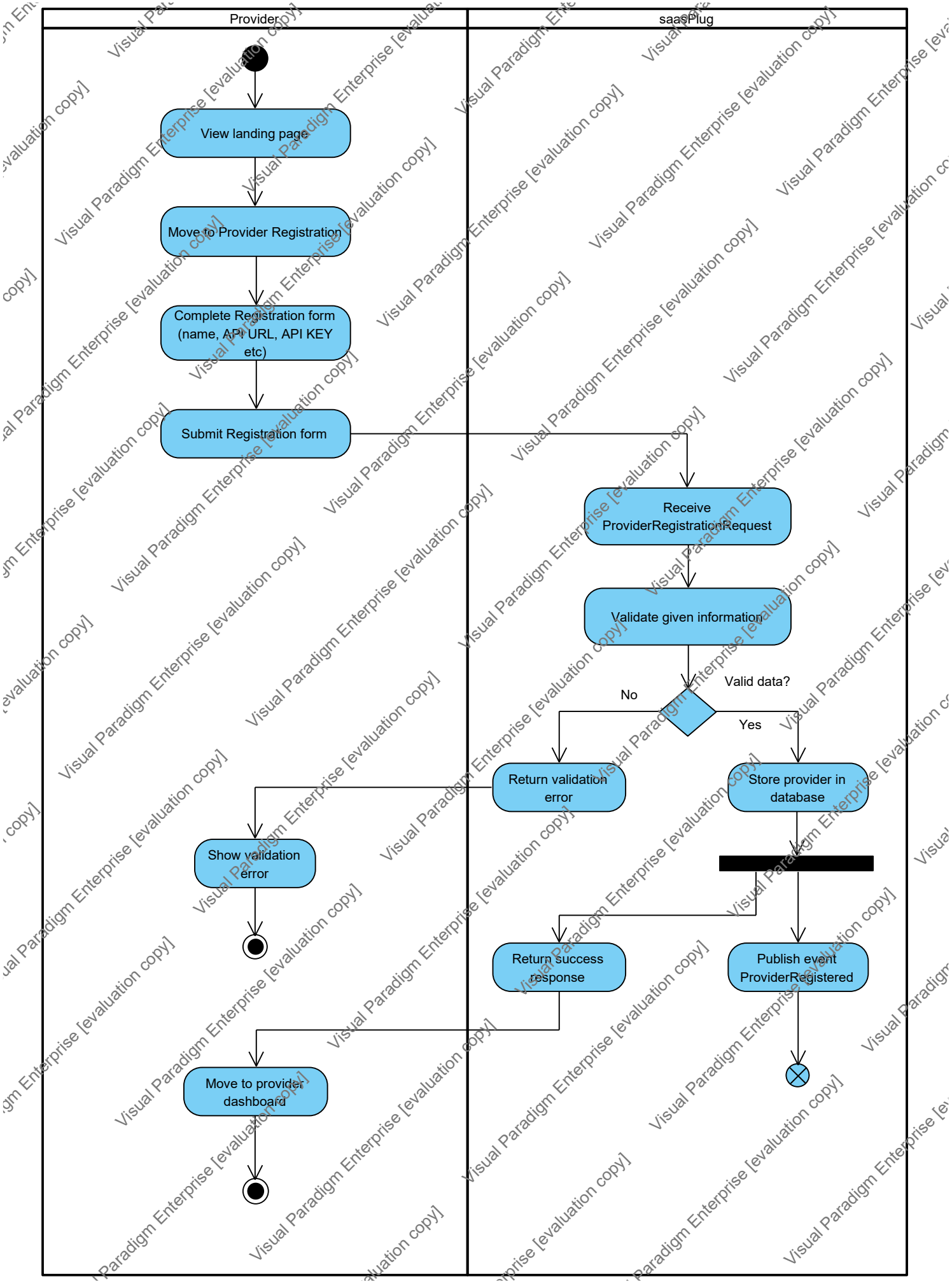

EV User
React Web App : Boundary
Nginx Gateway : Gateway
Reservation Service : Microservice
Charger Service : Microservice
Plug Service : Microservice
Analytics Service : Microservice
1: selectCharger(chargerId, providerName)
2: setDuration(minutes) + clickReserve()
2.1: POST /api/reservations
alt
charger status != "available"]
3: HTTP 409 Conflict
3.1: HTTP 409
3.1.1: Display error message "Charger Unavailable"
active reservation exists for userId]
4: HTTP 409 Conflict
4.1: HTTP 409
4.1.1: Display error message "Double Booking"
else]
5: Create Reservation
5.1: Persist reservation in database
5.1.1: HTTP 201 Created
5.1.1.1: HTTP 201 Created
5.1.1.1.1: Navigate to Charging screen
6: Update charger status = reserved
7: Call provider adapter
7.1: Route to correct adapter
8: Increment reservation statistics
9: clickCancel()
9.1: POST /api/reservations/{id}/cancel
9.1.1: Forward request with JWT authentication
9.1.1.1: Update reservation status = cancelled
9.1.1.1.1: Update charger status = available
10: HTTP 200 OK
10.1: HTTP 200 OK
10.1.1: Reservation cancelled successfully

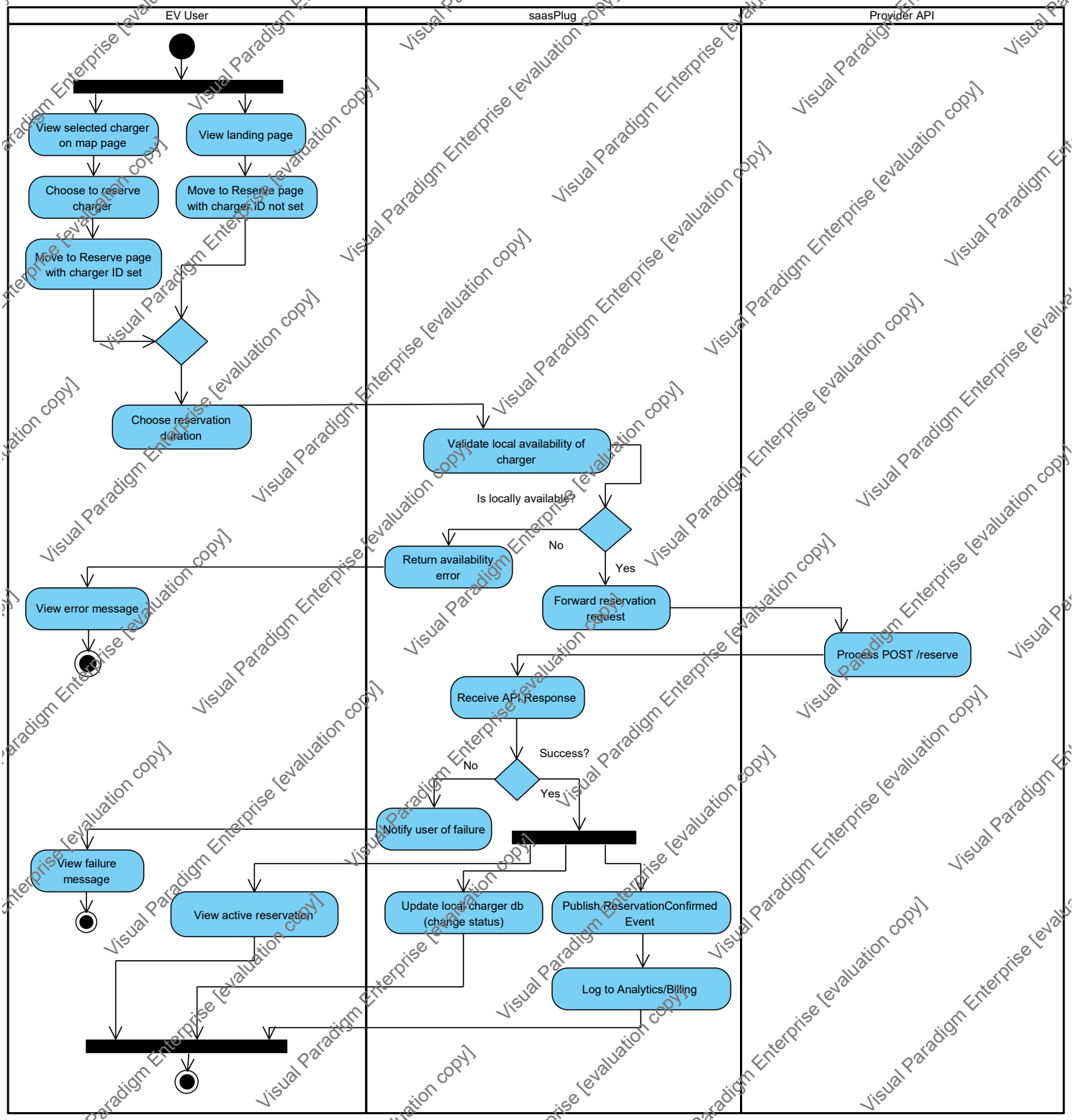

Admin
React Web App : WebApp
nginx API Gateway : Gateway
Admin Service : Service
Analytics Service : Service
alt
1: Maps to /admin — Stats tab
1.1: GET /api/admin/analytics
1.1.1: forward + admin JWT
1.1.1.1: GET /analytics/global
[[Apply Filter]]
2: select provider / date filter
2.1: GET /api/admin/analytics?provider=X
2.1.1: forward + admin JWT
2.1.1.1: GET /analytics/providers (filtered)
[[Generate Invoice]]
3: select provider + month + clickGenerateInvoice()
3.1: POST /api/analytics/invoices/generate {providerName, period}
3.1.1: forward + admin JWT
3.1.1.1: totalRevenue x 10% + sessions x 0.30 EUR
3.1.1.1.1: create Invoice (status: pending)
[[Mark Invoice Paid]]
4: clickMarkPaid(invoiceId)
4.1: PATCH /api/analytics/invoices/:id/paid
4.1.1: forward + admin JWT
4.1.1.1: status = paid

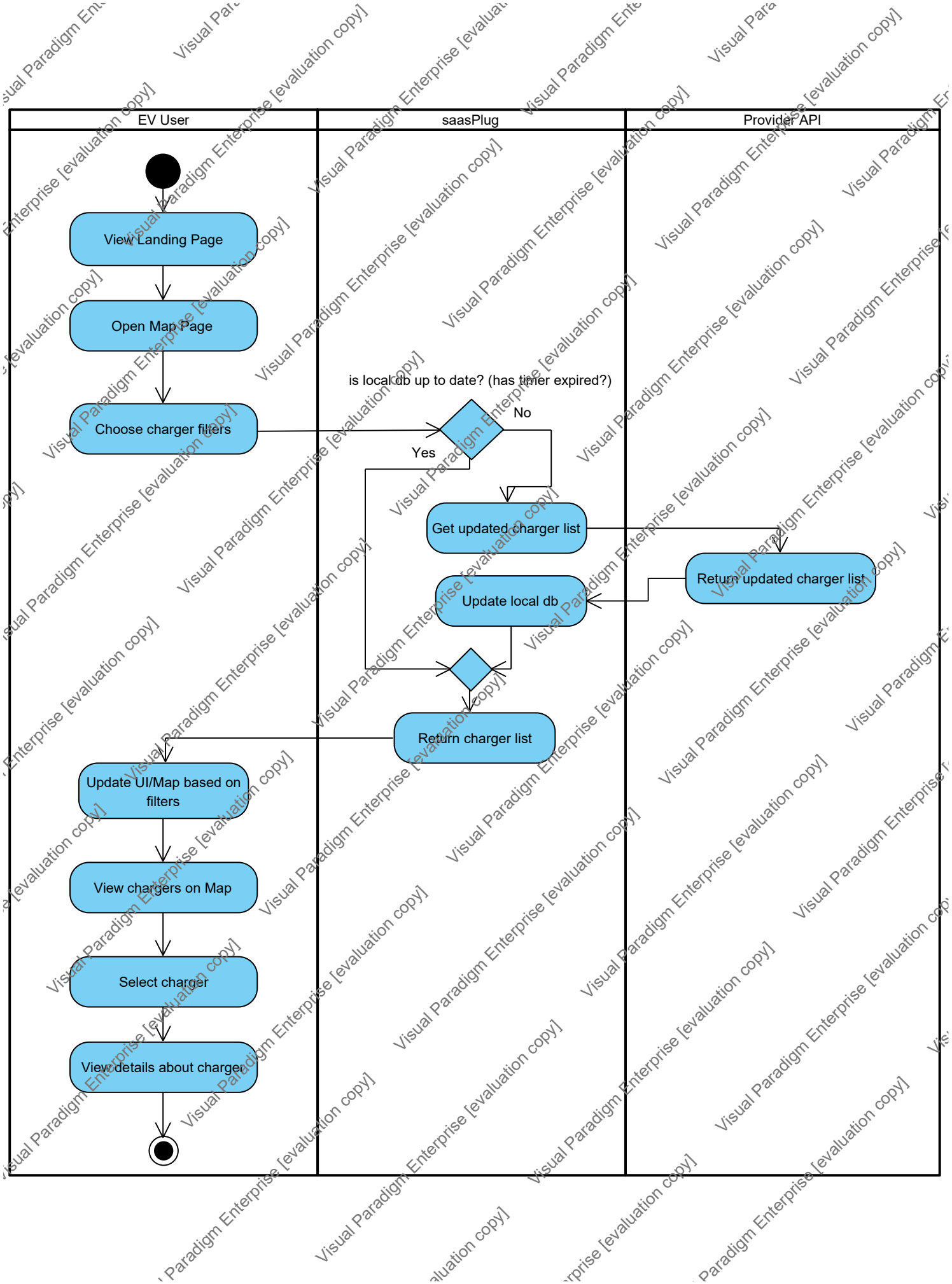

alt
[else]
Provider
React Web App : WebApp
nginx API Gateway : Gateway
Analytics Service : Analytics
1: navigate to /provider (dashboard)
1.1: GET /api/analytics/provider/:providerName
1.1.1: forward + provider JWT
[JWT rejected]
2: 401 Unauthorized
2.1: redirect to login
[else]
3: verify JWT role = provider
3.1: match providerName from token
3.1.1: aggregate data for provider
3.1.1.1: AnalyticsReport {providerName, period, totalChargers, chargersByStatus, utilizationRate, averageKwhPrice, totalReservations, averageDurationMinutes, totalRevenue, totalEnergyKwh}
3.1.1.1.1: dashboard (stat cards + charger fleet table)
{}
[Apply Period Filter]
4: change period (YYYY-MM)
4.1: GET /api/analytics/provider/:providerName?period=YYYY-MM
4.1.1: forward + provider JWT
4.1.1.1: query with period filter
4.1.1.1.1: filtered AnalyticsReport
4.1.1.1.1.1: re-render with filtered data

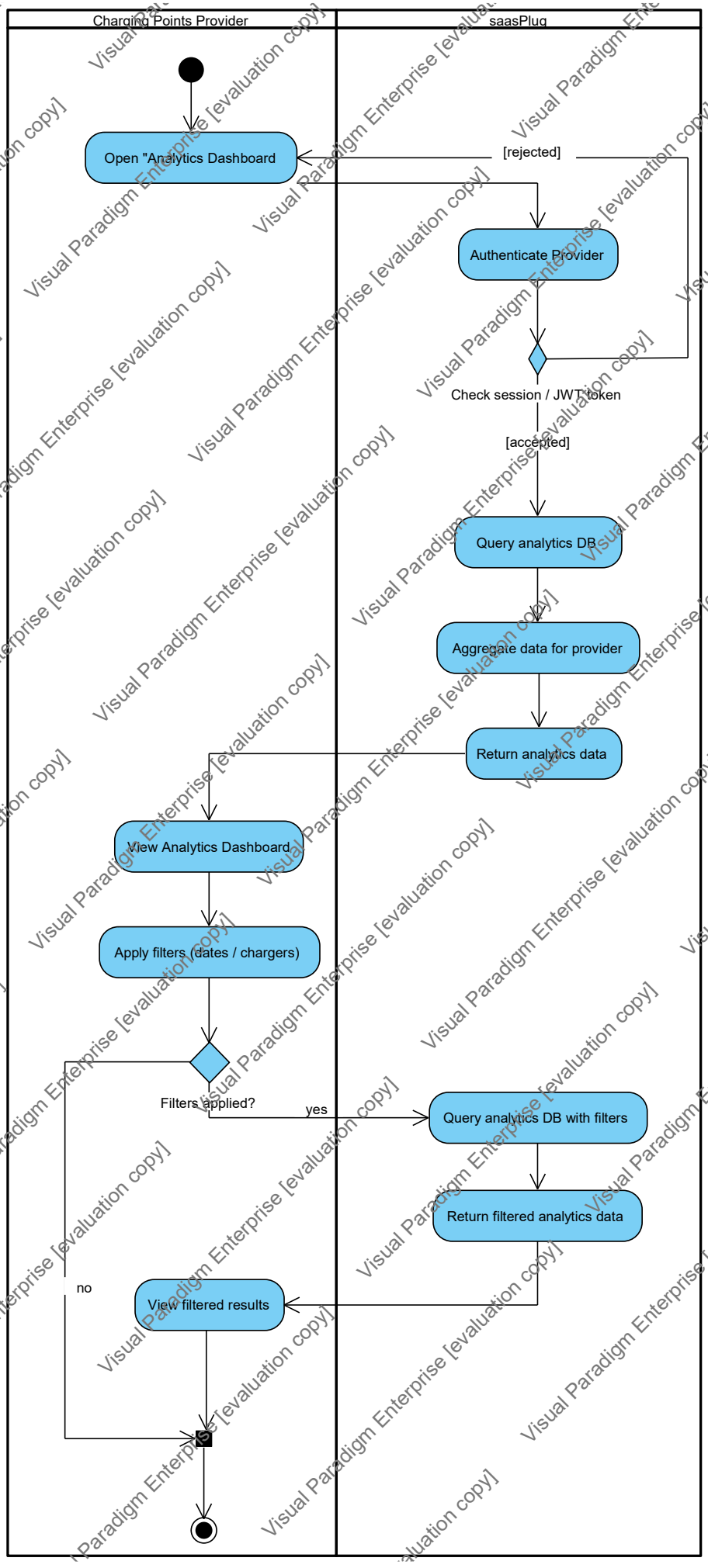

Provider
React Web App : Component
nginx API Gateway : Component
Analytics Service : Component
Reservation Service : Component
1: Maps to /provider/invoices
1.1: GET /api/analytics/invoices/my
1.1.1: forward + provider JWT
alt
provider JWT exists]
1.1.1.1: filter by providerName from JWT
1.1.1.1.1: invoice
1.1.1.1.1.1: invoice table (period, sessions, revenue, totalDue, status)
else]
2: no provider credentials found
3: clickViewInvoice(invoiceId)
3.1: GET /api/analytics/invoices/:id/detail
3.1.1: forward + provider JWT
3.1.1.1: GET /reservations/internal/by-provider?providerName=X&period=Y
3.1.1.1.1: saasPlugReservationResponse
3.1.1.1.1.1: build ReservationLine
3.1.1.1.1.1.1: InvoiceDetail {invoice, lines

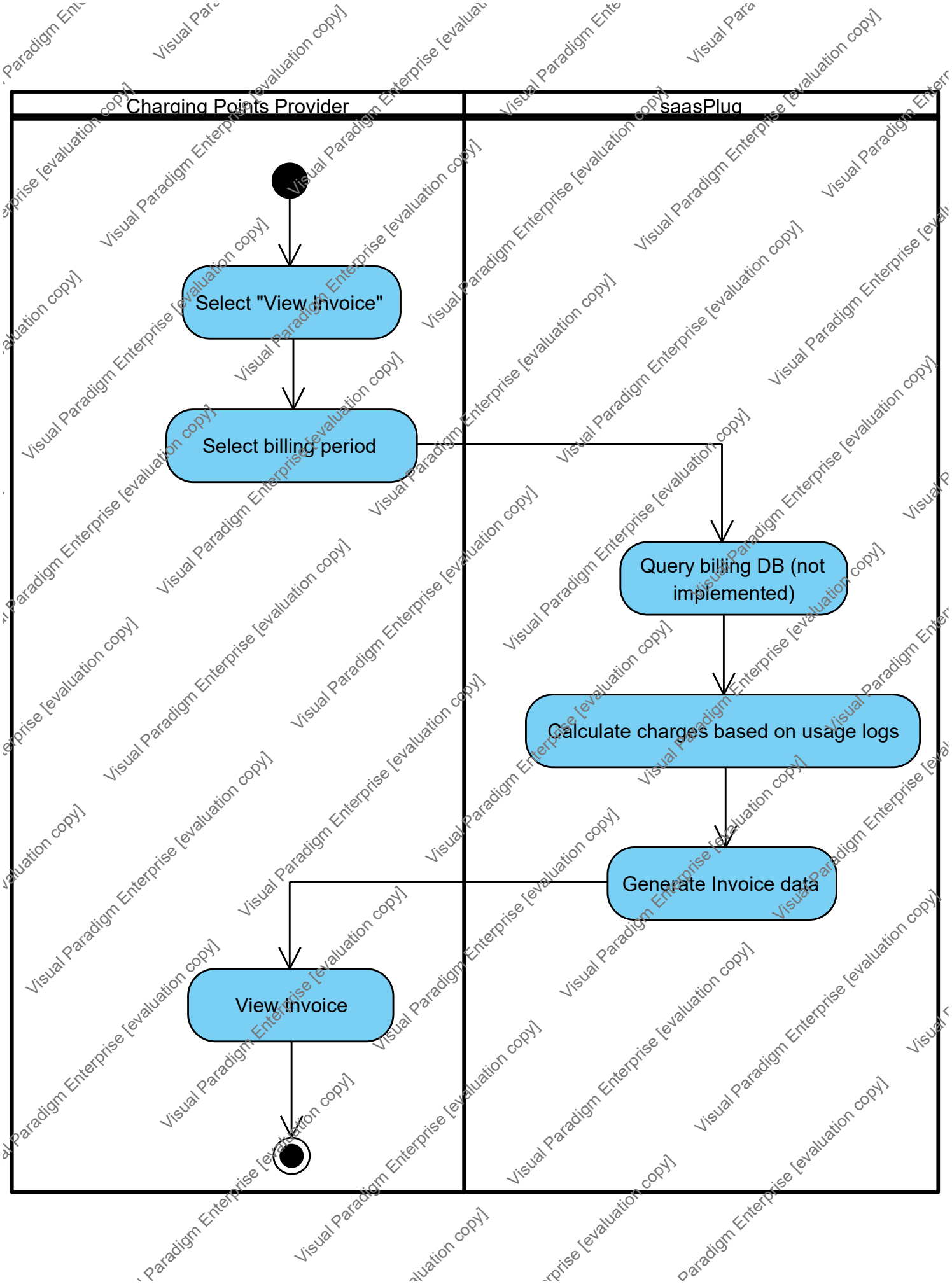

saasPlug operator
saasPlug
Open Global Analytics dashboard
Authenticate
Verify Operator credentials
Query analytics DB (all providers)
Aggregate global usage data
Return global analytics data
View global analytics data
Apply filters (provider / date / charger)
Filters applied?
yes
no
Query analytics DB with filters
Return filtered data
View filtered results


Provider
saasPlug
View landing page
Move to Provider Registration
Complete Registration form (name, API URL, API KEY etc)
Submit Registration form
Receive ProviderRegistrationRequest
Validate given information
Valid data?
No
Yes
Return validation error
Store provider in database
Show validation error
Return success response
Publish event ProviderRegistered
Move to provider dashboard


EV User
saasPlug
Provider API
View selected charger on map page
View landing page
Choose to reserve charger
Move to Reserve page with charger ID not set
Move to Reserve page with charger ID set
Choose reservation duration
Validate local availability of charger
Is locally available?
No
Yes
Return availability error
View error message
Forward reservation request
Process POST /reserve
Receive API Response
Success?
No
Yes
Notify user of failure
View failure message
View active reservation
Update local charger db (charge status)
Publish ReservationConfirmed Event
Log to Analytics/Billing


EV User
saasPlug
Provider API
View Landing Page
Open Map Page
Choose charger filters
is local db up to date? (has timer expired?)
No
Yes
Get updated charger list
Return updated charger list
Update local db
Return charger list
Update UI/Map based on filters
View chargers on Map
Select charger
View details about charger


Charging Points Provider
saasPlug
Open "Analytics Dashboard
[rejected]
Authenticate Provider
Check session / JWT token
[accepted]
Query analytics DB
Aggregate data for provider
Return analytics data
View Analytics Dashboard
Apply filters (dates / chargers)
Filters applied?
yes
Query analytics DB with filters
Return filtered analytics data
no
View filtered results


Charging Points Provider
saasPlug
Select "View Invoice"
Select billing period
Query billing DB (not implemented)
Calculate charges based on usage logs
Generate Invoice data
View Invoice